BT404 (Food Biotechnology)

Final Term (Current Papers 2022)

Solved By ~ Bella

We Grow Together

Cereal grain composed of what parts?

Most common cereal foods include:

- Rice
- Wheat
- Maize (corn)

Millets such as:

- Jowar
- Bajra
- Ragi

Can be processed further to make:

- Flour
- Starch
- Oil
- Bran
- Syrup
- Sugar

Economics of evaporation?

Economy (or steam economy) is the kilograms of water vaporized from all the effects (per kilogram of steam used). For single effect evaporator, the steam economy is ca. 0.8 (<1), which translates to 0.8 tons of steam needed to evaporate 1 ton of water.

Which Factors influencing the economics of evaporation?

- Vapour recompression
- Preheating
- Multiple effect evaporation

How lipids are classified?

Lipids are classified as

- Simple lipids
- Complex lipids

- Derived lipids

Function of starch synthases?

- Starch synthases (SSs) are responsible for depositing the majority of glucoses in starch.
- Catalyze the polymerization of the glucose monomers into alpha-1,4-glucans
- Using ADP glucose as a substrate
- Starch branching enzyme catalyzes the formation of the alpha-1,6-linkages of amylopectin

Function of SO2 in food preservation?

Sulphur dioxide can inhibit spoilage caused by microbes and extend the shelf life of food products. Other than its antimicrobial properties, sulphur dioxide is also an antioxidant that prevents browning associated with oxidation in foods, particularly dried fruits and vegetables.

Types of corn grown?

Many types of corn are grown:

- Sweet corn
- Popcorn
- Dent corn
- Flour corn

Note on Ultrasound? 5 M

Ultrasound :

It is used in food industry for many purposes including analysis methods and food processing such as:

- Freezing , Cutting
- Drying , Tempering
- Homogenization, degassing
- Antifoaming, filtration and extraction

Uses of Ultrasound :

Ultrasound can be used as a promoter or alternative to food processing.

Prevention Causes :

Ultrasound in food preservation causes:

- Microbial inactivation
- Enzyme inactivation

Selection of functional strain 5marks

Five key criteria for the selection of an appropriate functional strain:

1. General aspects
2. Origin
3. Identity
4. Resistance to mutations
5. Technical aspects (growth properties in vitro and during processing, survival and viability during transport and storage).

Calculation of titratable acidity? 5marks

% acid (wt/wt) = [N x V x Eq wt / W x 1000] x 100

N= normality of titrant

V= volume of titrant

Eq wt = equivalent weight of predominant acid

W = mass of sample

1000 = factor (mg/g)

Write the Importance of Water in Foods?

- Water as Quality Factor in Preservation such as in products like dried milk, dehydrated vegetables, spices and herbs.
- Quality factor of products Such as jams, jellies, fruit syrups, prepared cereals etc.
- Convenience in Transportation reduced moisture food is helpful in packaging and shipping such as concentrated milk, liquid cane sugar, concentrated fruits juices.
- Moisture as compositional standards Such as in foods cheese, flour, pineapple, meat etc.
- Moisture as nutritional Standards
- Moisture data as expressed in analytical results of many foods such as food contents on wet basis and dry basis.

What is Cryogenic refrigerator?

Cryogenic Refrigerators

A cryogen is a refrigerant that changes phase by absorbing latent heat to cool the food. Cryogenic chillers use solid carbon dioxide, liquid carbon dioxide or liquid nitrogen. Solid carbon dioxide removes latent heat of sublimation and liquid cryogens remove latent heat of vaporization.

Note on probiotics?

Probiotics are live bacteria and yeasts that are good for you, especially your digestive system. We usually think of these as germs that cause diseases. But your body is full of bacteria, both good and bad. Probiotics are often called "good" or "helpful" bacteria because they help keep your gut healthy.

Types of Probiotics

- ✓ Lactobacillus
- ✓ Bifidobacterium
- ✓ Saccharomyces boulardii

Probiotics are milk products containing active bacteria like Lactobacillus acidophilus, Lactobacillus casei, Bifidobacterium bifidium, etc. These bacteria maintain the balance of intestinal micro- organisms by increasing the population of microbes that help in digestion and decreasing the population of harmful microbes such as Clostridium. They also improve resistance and lower the ill effects of harmful substances formed during metabolic activities. When we take antibiotics, the useful microbes in the alimentary canal become inactive. These microbes can be made active by consuming probiotics. Also, probiotics are used for the treatment of diarrhoea and treatment of poultry diseases.

Equation of ash percentage?

$$\%\ \text{ash (dry basis)} = \frac{\text{wt after ashing} - \text{tare wt of crucible}}{\text{original sample wt} \times \text{dry matter coefficient}} \times 100$$

Names of Sweetening agents?

Most commonly used sweeteners:

- Table sugar (sucrose)
- Corn Syrup

Many sweeteners are classified as non-nutritive

Fructose

- Molasses
- Honey
- Maple sugar
- Lactose
- Maltose
- Polyhydric alcohols
- Aspartame
- Saccharin

Natural

- Monosaccharide twice as sweet as sucrose
- Most water soluble
- Hygroscopic

Write the 3 issues in hygiene of food?

- Changes in our food production and supply, including more imported foods.
- Changes in the environment leading to food contamination.
- Better detection of multistate outbreaks.
- New and emerging bacteria, toxins, and antibiotic resistance.
- Changes in consumer preferences and habits.
- Changes in the tests that diagnose foodborne illness.

Write the main issues in hygienic design of food processing?

The main issues of hygienic design are:

- To avoid contamination by foreign organisms and materials like animals, insects, birds, microorganism, migrations, construction/ equipment parts, lubricants , residues of other products and cleaning agents, etc.
- To avoid conditions which enhance the growth of microorganism like accumulation of dust, water, moisture (condensation) and product
- To improve cleanability

Temperature of incubation for microorganisms?

37 degrees C for 48 h on suitable culture media.

Name of two animal toxin?

- Tetrodotoxin
- Scrombridae

Name of enzymes reagent use in cheese?

Rennet includes the enzyme chymosin to help curdle the cheese.

Which factor can effect the quality of food?

- ✓ Appearance factors
- ✓ Textural factors
- ✓ Flavour fact**ors**.

Use of Refractometer in food?

The refractometer provides an alternative means of measuring the soluble solids of sugar syrups and fruit products and is based on the index of refraction of the solution.

Types of botulism?

There are five types of botulism:

- Food-borne botulism
- Wound botulism
- Infant botulism
- Adult infectious botulism
- Inadvertent botulism

What is Agar plating?

Plating surface spread-plate technique rather than with pour plates for maximal exposure of the cells to atmospheric oxygen and to avoids heat stress from molten agar. Agar spread plates should be dried overnight before being inoculated.

Briefly note on Vitamin E?

Vitamin E has four different forms (the tocopherols) that have the same name except with the prefixes alpha-, beta-, gamma-, and delta-. It serves as an antioxidant that serves to prevent the oxidation of body components. Good sources of Vitamin E are corn oil, cottonseed oil, and peanut oil. Deficiency effects reproduction, injury to the central nervous system, growth retardation, muscular dystrophy, and interference with normal heart action. The RDA for vitamin E is now given in milligrams of alpha-tocopherols equivalents (a-TE).

Hydrogenation in oil and fat?

Hydrogenation oils are exposed to hydrogen gas under conditions of high temperature (150–180◦C) and pressure (2–10 atm) in the presence of a nickel catalyst.

Safety factor for quality of cheese making?

- ✓ Centrifugation
- ✓ Standardization
- ✓ Homogenization
- ✓ Pasteurization

Names of culture use in cheese?

Types of cultures :

- ✓ Liquid (for propagation of mother culture; rarely used today)
- ✓ Deep-frozen concentrated cultures (for propagation of bulk starter)
- ✓ Freeze-dried concentrated cultures in powder form (for propagation of bulk starter or DRI-VAC, for preparation of mother culture)
- ✓ Deep-frozen or freeze-dried, super concentrated cultures in readily soluble form for direct inoculation of the product (direct vat set, DVS).

Function of sodium hydroxide and calcium propionate in food preservatives?

Sodium Hydroxide :

Sodium hydroxide is used to remove skins from tomatoes, potatoes and other fruits and vegetables for canning and also as an ingredient in food preservatives that help prevent mold and bacteria from growing in food.

Calcium Propionate :

Calcium propionate is an organic salt that is used as a food additive. It helps preserve foods, mainly baked goods, by interfering with the growth and reproduction of microorganisms, such as molds, bacteria, and fungi.

What is Retrogradation?

Retrogradation is a reaction that takes place when the amylose and amylopectin chains in cooked, gelatinized starch re-align themselves as the cooked starch cools.

Four general classes of Sorghum?

- ✓ Sweet sorghum
- ✓ Broom corn
- ✓ Grass sorghum
- ✓ Grain sorghum

How probiotics are administered?

Probiotics may be administered in different forms

- ✓ Traditional fermented foods
- ✓ Food products
- ✓ Pharmaceuticals

Cheese Milk Pre-treatments?

- Centrifugation
- Standardization
- Homogenization
- Pasteurization

What are examples of food regulations that address food safety and food quality?

- U.S. Federal Food, Drug, and Cosmetic Act (FDCA), which is the primary law governing the safety and quality of most foods in the U.S.
- Canada's Food and Drugs Act, which is the primary food law in Canada.
- The U.S. Code of Federal Regulations (CFR) Title 21 and Canada's Food and Drug Regulations are examples of food regulations that address food safety and food quality.

The main reasons for food preservation?

- Food preservation gives the food more variety.
- Food preservation extends food's shelf-life.
- Food preservation expands the supply of food.
- Food preservation cuts down on food waste.
- Food preservation helps to reduce dietary deficiencies. Preserved foods help to add variety to the diet.

Seven Principles of HACCP?

These seven principles are

1. Hazard analysis
2. Critical control point identification
3. Establishment of critical limits
4. Monitoring procedures
5. Established corrective actions
6. Record keeping
7. Verification procedures.

Food Safety Hazards?

There are three recognized categories of food safety hazards:

Biological hazards

Biological hazards include microorganisms such as bacteria, viruses, yeasts, molds and parasites. Some of these are pathogens or may produce toxins.

Chemical hazards

Chemical hazards include water, food contact materials, cleaning agents, pest control substances, contaminants (environmental, agricultural and process e.g. acrylamide), pesticides, biocides and food additives.

Physical hazards

Physical hazards include objects that are hard or sharp such as glass, metal, plastic, stones, pits, wood, or even bone. Physical hazards can lead to injuries such as choking, cuts, or broken teeth.

Limitations of UHT?

The main limitations of UHT processing are the

- Cost
- Complexity of the plant which arises from the necessity to sterilise packaging materials and pipework/tanks
- Maintenance of sterile air and surfaces in filling machines.
- Higher skill levels required by operators and staff

Vegetables names horboric lactic acid bacteria? 5M

Almost all vegetables can be fermented through natural processes because they harbour many types of lactic acid bacteria originated in the early years of human civilization and even now is widely used by many cultures.

Names of Vegetables

Sauerkraut (from cabbage), Olives, cucumbers, carrots, celery, beans, peas, corn, okra, tomatoes, cauliflower, peppers, onions, citron, beets, turnips, radishes, chard, Brussels sprouts, and their blends.

Write the selective temperature for the incubation of microorganisms?

Use of Selective incubation temperature for isolation :

- 55-65 °C (Thermophiles)
- 20 °C (Physchrophiles)

Which chemical are responsible for fruity odour?

Esters of low molecular weight are responsible for fruity odours and consist of acids and their derived compounds such as acetates, propionates, and butyrates. Some examples are ethyl butyrate and isoamyl acetate, which are found in strawberry and banana .

Note on glucose oxidase and detection of amylase enzyme?

Glucose oxidase :

Glucose oxidase is an enzyme that specifically catalyzes the oxidation of glucose to gluconic acid. Non enzymatic browning, is a undesirable browning reaction. The most important application of this enzyme is in the treatment of egg products, especially egg whites, prior to drying. In some cases, the enzyme is added to remove traces of oxygen to prevent oxidative degradation of quality. Examples of this type of application are mayonnaise and bottled and canned beverages (especially beer and citrus drinks).

Detection of amylase of enzyme :

Amylase activity is determined using a coupled enzymatic assay, which results in a colorimetric (405 nm) product, proportional to the amount of substrate, ethylidene-pNP-G7, cleaved by the amylase.

Name of bacteria in yogurt?

Lactic Acid Bacteria (LAB)

Why are fats used in food ?

The body uses fat as a fuel source, and fat is the major storage form of energy in the body. Fat also has many other important functions in the body, and a moderate amount is needed in the diet for good health.

Write a note on Listeria monocytogens?

Listeria monocytogens: are Gram-positive, motile bacteria. They do not form spores, but are quite resistant to desiccation and heat. Found in the soil and associated with birds and animals. Cause disease called listeriosis, which can result in septicaemia, meningitis encephalitis and spontaneous abortions in pregnant females.

What are the functions of food substances when ingesting by living organisms?

Foods substances that are ingested by living organisms and can be metabolized to perform following functions:

- Source of energy
- Source of raw materials for growth and development
- To supply minute amount of nutrients to regulate vital metabolic processes

- To supply food components (phytochemicals) that retard the development and progression of degenerative diseases.

Foods can be fermented in three different ways based on the sources of the desirable microorganisms. What are they?

- Natural fermentation
- Back slopping
- Controlled fermentation

Define Dietary fiber?

Dietary fiber or roughage is the portion of plant-derived food that cannot be completely broken down by human digestive enzymes. Dietary fiber includes the non digestible carbohydrates. These may be either water soluble or water-insoluble. The water insoluble group, which includes wheat products and wheat bran. The water-soluble fibers, includes such as some cereal brans and pectin.

Mention at least three names of Microbes that pasteurization inhibits the growth?

- Bacillus
- Micrococci
- Lactococci
- Arthrobacter
- Microbacteria
- Corynebacteria

What are the types of rice ?

Rice is available as

- Milled
- Unpolished
- Polished
- Parboiled
- Faked
- Puffed

Write down the steps of Refining of crude oils?

Refining is a 5 step process

- Degumming
- Neutralising
- Washing and Drying
- Bleaching
- Deodourising

How infra red dying process is done?

Infra red lamps with a filament temperature of 2000-2500 K can penetrate heat to food and shorten drying time to 20-25 minutes only.

What is phenylethylamine?

- Found in chocolate
- Vasopressor amine
- Constrict blood vessels
- Can cause migraine in susceptible person.

What are the Heat control methods to control Microbial growth?

Heat control methods to control Microbial growth :

High Temperature :

- Pasteurization
- Sterilization
- UHT etc

Low Temperature :

- Blast freezing
- Cryogenic freezing etc.

What is the Ideal Solvent For Fat Extraction?

- High solvent power for lipids
- Low solvent power for nonlipids
- No residue
- Evaporate easily
- Low boiling point
- Non flammable / not explosive
- Nontoxic
- Low surface tension with food
- Single compound
- Non-hygroscopic

What are the Ingredients used for Cheese Making?

- Milk
- Calcium Chloride
- Starter Culture
- Adjunct Culture
- Colour
- Enzyme
- Salt
- Optional ingredients

Write two types of wheat ?

There are two types of wheat:

- Hard wheat (High Protein)
- Soft wheat (Low Protein)

Write the used products of wheat ?

Common used products of wheat

- Vermicelli
- Broken wheat (Dalia)
- Noodles
- Suji
- Cereals
- Canned soups
- Pasta

Which factors affecting Microbial growth?

Factor affecting Microbial growth are

- pH
- Moisture
- Humidity of environment
- Temperature
- Availability of Oxygen

Sequence of Events in Food Borne Diseases ?

A sequence of events happens in all food borne diseases. An understanding of these sequences is helpful in investigating the cause (the source and means of transmission) of a food borne disease. It also helps in recognizing how the sequence can be broken.

- Initially, there has to be a source of a pathogen.
- The pathogen has to contaminate a food.
- Consumption of the food contaminated with a pathogenic virus or parasites.
- It lead to viral or parasitic infection.

Define foodborne disease?

Foodborne disease is caused by consuming contaminated foods or beverages. Many different disease-causing microbes or pathogens can contaminate foods, so there are many different types of foodborne diseases. Most foodborne diseases are infections caused by a variety of bacteria, viruses, and parasites.

What is normal range of protein and fats in meat?

The average protein content of meat ranges from 16 to 23 % and the average fat content ranges from 10 to 40%.

Name any two optional ingredients used for cheese making?

- Sodium benzoate
- Sodium propionate
- Sorbic acid
- Hydrogen peroxide

Define plasticity of Fats?

Fats are mouldable and can be creamed to exhibit plasticity. Such fats do not flow at room temperature and are thus solid fats. The spreading quality of butter is the result of its plastic nature. Plastic fats are composed of a mixture of triglycerides and not of one kind of a molecule.

What is Oxidizing agents?

An oxidizing agent, also known as an oxidant or oxidizer, is a substance that has the ability to oxidize other substances. Common oxidizing agents are oxygen, hydrogen peroxide and the halogens.

What are cereals products are made from grains?

A number of processed cereal products are made from grains Include products like:

- ✓ Cracked wheat
- ✓ Semolina
- ✓ Atta
- ✓ Maida
- ✓ Rice flakes
- ✓ Puffed rice

Steps of cleaning and sanitation in food processing?

- ✓ Remove Debris
- ✓ Rinse All Residues
- ✓ Apply Detergent and a Good Scrubbing
- ✓ Give a Thorough Rinse
- ✓ Take a Closer Look
- ✓ Sanitize or Disinfect
- ✓ Dry

Types of lipids?

- ✓ Fatty acids (Saturated or Unsaturated)
- ✓ Glycerides (Glycerol containing lipids)
- ✓ Non- Glyceride lipids (Sphingolipids, Steroids or Waxes)
- ✓ Complex lipids (Lipoproteins , Glycolipids)

Five baked products ?

- ✓ Biscuits
- ✓ Cakes
- ✓ Bread

- ✓ Pizza
- ✓ Doughnuts
- ✓ Muffins
- ✓ Pie

Overview of marketing?

Marketing refers to the activities pursued by a company to promote its products or services to the target market. The goal of marketing is to draw the consumers' attention to the company's products and create a relationship with existing customers.

Application of proteolytic enzyme in food industry? 3 Marks

In the manufacture of certain kinds milk powder (e.g., to be used in chocolate), the lipases may be allowed to act on the milk prior to drying to obtain a particular flavour in the finished product. Proteases may also be used to chill-proof beers, a technique used to remove proteins that would cause clouding during cool storage.

Oxidizing enzymes in food biotechnology? 5 Marks

Oxidizing enzymes that bring about changes in foods that result in deterioration due to oxidation. In plants Peroxidases, ascorbic acid oxidase, tyrosinase, and polyphenolases may cause Undesirable chemical reactions to occur. Peroxidases may oxidize certain phenol like compounds in root vegetables, such as horseradish, causing the prepared product to become darker in colour when damaged. Ascorbic acid oxidase, in vegetables, oxidizes ascorbic acid cause loss of the vitamin C content of foods. Peroxidases also, indirectly, cause a loss of vitamin C by its by-products. Phenolases are present in some fruits and vegetables cause brown or dark-coloured compounds to be formed when the tissues are cut.

What is homogenization milk ?

Most efficient when fat globules are in the liquid state ,so milk is preheated in the plate heat exchanger in the regeneration section of the HTST pasteurizer, where the temperature is raised to at least 60C prior to homogenization. In a two-stage homogenizer, pressures typically range from 10 to 25 MPa in the first stage and 5 MPa in the second stage.

What is Edible vaccines? 2 Marks

Edible vaccines like traditional subunit vaccines consist of antigenic proteins and are devoid of pathogenic genes. Edible vaccines posses good genetics , Heat stability and do not need cold-chain maintenance, can be stored at the site.

What are triglycerides? 2 Marks

Fats and oils are mixture of different triglycerides Triglycerides are formed by mixing of glycerol and fatty acids with elimination of a water molecule.

Define the term entrapment type bioreactors in scientific words? 2 Marks

Entrapment type is very common bioreactor, where high concentrations of cells are trapped in a polymer matrix. It has high cell density is continuously retained in the fermenter continuously converting substrate to product. This results in higher productivities and yield.

The conveying capacity of a bucket elevator depends upon which factors? 3 Marks

- ✓ Product density
- ✓ Belt speed
- ✓ Bucket size
- ✓ Spacing of the bucket in the belt

Name food resources which contain stone? 2 Marks

Many plant foods and particularly field crops such as peas and beans can contain small stones that become incorporated with the foods during harvesting.

What are the uses of Gums? 3 Marks

- ✓ Stabilize ice-cream and desserts
- ✓ Thicken beverages and preserves
- ✓ Stabilize foam in beer
- ✓ Emulsify salad dressing
- ✓ Form protective coatings from meat and fish

Name agar which detects the yeast? 5 Marks

- ✓ Dichloran rose Bengalchloramphenicol agar (DRBC)
- ✓ Plate count agar with chloramphenicol (PCAC)
- ✓ Potato dextrose agar with chloramphenicol (PDA-C)
- ✓ Potato dextrose salt agar with chloramphenicol (PDSA-C) (for analysis of spreader moulds)

Name two bacteria which used in dairy product?

- ✓ Lactobacilli
- ✓ Bifidobacteria

What are catalases? 2 Marks

Catalases are used to break down hydrogen peroxide (H202) to water and oxygen. Catalases are used when the presence of the hydrogen peroxide is undesirable or when hydrogen peroxide is used for specific purposes, such as in bleaching, but then must be removed from the system.

What are unique characteristics of cottage cheese? 3 Marks

Cottage Cheese:

- ✓ From low-fat or skim milk and has a soft texture with Ca. 80% moisture.

- ✓ It is unripen and has a buttery aroma due to diacetyl (along with lactic acid and little acetaldehyde).
- ✓ Made from whole milk contains less than 39% moisture, 48% fat.
- ✓ Starters can be used as frozen concentrates.

What are the steps of Buyer process? 5 Marks

Following eight steps are suggested to constitute the buyer decision process:

- ✓ Recognize problems
- ✓ Specify the productions
- ✓ Seek suppliers
- ✓ Solicit proposals
- ✓ Select the supplier
- ✓ Order the desired specification
- ✓ Review the performance

What are updated seven Ps of marketing? 3 Mark

The 4P's have been updated to now 7P's

- ✓ Product
- ✓ Price
- ✓ Place
- ✓ Promotion
- ✓ People
- ✓ Positioning
- ✓ Packaging

What is buttery honey or toffee like flavour? 2 Marks

Diacetyl and pentane-2,3- dione (vicinal diketones) have a characteristics flavour described as "buttery" “honey” or "toffee- like". They have a very high off-flavour potential dependent on the fermentation temperature.

How microbes effect fermented vegetables product? 2 Marks

Off-flavour, soft texture, and discoloration of vegetables can occur by growth of molds and yeasts when air is not completely excluded.

What are conjugated proteins?

Proteins are also sometimes combined with carbohydrates, lipids, or phospholipids called as conjugated proteins.

Define gums? Also give its examples?

Gums :

Those materials that are dispersible in water and make the water more viscous are called gums.

Examples of Gums :

- ✓ Gum Arabic
- ✓ Agar
- ✓ Vinyl polymer gums are synthetic
- ✓ Cellulose derivate gums are modified or semi synthetic

What are the common defects in cheeses?

Common flavour defects are high acid, bitter, unclean, and fermented /fruity.

Common body defects are weak or crumbly body, gas holes, surface discoloration, and appearance of crystals on surfaces.

Common texture defects may be corky, crumbly, crystals, meaty, gassy, open, pasty, short and weak

What are the oxidative quality changes in foods?

The development of rancidity from the oxidation of unsaturated fats resulting in off-odours and off-flavours and discoloration from oxidation of pigments or other components of the food.

Write down the Theories of the flavours?

First theory is that they increase the sensitivity of the taste buds, thus increasing flavour.

A **second theory** suggests that an increase in salivation as a result of the flavour enhancers.

A **third theory** of intensified flavour perception is flavour enhancers produce certain physical sensations in the mouth such as coolness and heat.

How rendered fats can be refined?

- ✓ Sodium carbonate
- ✓ Sodium bicarbonate

Define potentiometric method?

A potentiometric method is used for deeply coloured samples.

What are the applicability of probiotics in food products depends on?

- ✓ Water activity
- ✓ Processing and storage temperature
- ✓ Shelf life
- ✓ Oxygen content
- ✓ pH
- ✓ Mechanical stress
- ✓ Salt content
- ✓ Content of other harmful or essential ingredients.

Write the names of the major probiotic microorganisms?

- ✓ Bacteroides
- ✓ Eubacterium
- ✓ Bifidobacterium

Write down the functions of normal gut bacteria?

- ✓ Maintenance and restoration of barrier function
- ✓ Stimulation of the immune system
- ✓ Maintenance of mucosa nutrition and circulation Improvement of bioavailability of nutrients
- ✓ Stimulation of bowel motility and reduction of constipation

Which four polyols are only allowed?

- ✓ Glycerine
- ✓ Sorbitol
- ✓ Mannitol
- ✓ Propylene glycol

In which type of production we used polyols?

Polyols are used in the production of dietetic products

- ✓ Beverages
- ✓ Candies
- ✓ Gums
- ✓ Ice creams

Write down the name of 2 indicator dyes that are used as end point?

- ✓ Phenolphthalein
- ✓ Bromothymol blue

Write the types of Colour additives?

Three major types

- ✓ Natural
- ✓ Nature-identical
- ✓ Synthetic

Why we use colour additives in foods?

Colour additives is used to make food more desirable and attractive.

Names of the organic compounds that affect the surface activities?

- ✓ Wetting agent
- ✓ Lubricant
- ✓ Dispersing agent
- ✓ Detergent

- ✓ Emulsifies

Names of inorganic salts for leavening?

- ✓ Ammonium salts
- ✓ Phosphate salts

Why a part of sugarcane is removed before harvesting?2 Marks

In the sugar cane, invertase breaks down sucrose, hence a part of sugarcane must be removed at harvesting.

Effect of heat on meat? 3 Marks

Upon heating meat changes greyish brown from the red pigment. As meat is heated the components that make it up change. These changes result in structural transformations that affect the texture, juiciness, and mouthfeel of the meat. The higher the temperature of the heat applied to the meat, the faster these changes happen.

Novel starches? 5 Marks

Novel, starch-like polysaccharides are synthesized by an unbound form of granule-bound starch synthase in glycogen. Starches obtained from different crops having varied functionalities due to differences in composition may now theoretically be readily transferred into crops from one or another different plant species. In general we can consider making changes in any aspect of amylose/amylopectin, phosphate, protein, phospholipids, crystallinity, gelation, and pasting characteristics, flavour and starch granule morphology.

Oatmeal nutrition? 5 Marks

Oats scientific name Avena Sativa. It is used as livestock feed. Oatmeal suitable for human consumption and oats most commonly rolled and crushed to make oatmeal and also grounded into flour. It is also used in variety of baked goods such as oatmeal cookies. Oats can grow in colder and wetter climates as compared to wheat or harvested in the same manner as wheat. Moisture content at time of harvest should be no more than 13%.Oats can be milled and may be produced for use as an ingredient of bread or thickener of soups. Oat bran has water soluble fiber and linked with lower cholesterol levels.

Strategy of food safety?

- ✓ Keep clean
- ✓ Separate raw and cooked
- ✓ Cook thoroughly
- ✓ Keep food at safe temperatures
- ✓ Use safe water and raw materials.

Two proteolytic enzymes in plants?

- ✓ Bromelain

- ✓ Papain

Briefly Note wheat crop?

Wheat is one of the oldest and most important of the cereal crops. Of the thousands of varieties known, the most important are common wheat (Triticum aestivum), used to make bread; durum wheat (T. durum), used in making pasta (alimentary pastes) such as spaghetti and macaroni; and club wheat (T. compactum), a softer type, used for cake, crackers, cookies, pastries, and flours. Additionally, some wheat is used by industry for the production of starch, paste, malt, dextrose, gluten, alcohol, and other products.

Starch to sugar balance?

Starch to sugar balance in which the overall change between the two compounds is seen as being reversible.

Kjeldahl Method to determine proteins involves two steps?

Kjeldahl Method to determine proteins involves these steps

- ✓ Sample digestion or wet oxidation
- ✓ Distillation of ammonia
- ✓ Titration of liberated ammonia

Unfermented pickle?

Unfermented pickles are made by marinating fruit or vegetables in a seasoned vinegar solution. They are also called Refrigerator pickles or Instant pickles. Unfermented pickles are produced by two methods:

- ✓ Salt-stock pickle
- ✓ Vinegared pickles”

What is MPN?

The most probable number (MPN) is a statistical method used to estimate the viable numbers of bacteria in a sample by inoculating broth in 10-fold dilutions and is based on the principle of extinction dilution. It is often used in estimating bacterial cells in water and food.

Define Nutrient additives?

Nutritional additives are used for the purpose of restoring nutrients lost or degraded during production, fortifying or enriching certain foods in order to correct dietary deficiencies, or adding nutrients to food substitutes.

Cheese making process?

Preparing the milk: Before it can be turned into cheese, the milk may need to be processed.

Acidifying the milk: Adding cultures to the milk allows it to begin to ferment and makes it more acidic.

Curdling the milk: Adding rennet causes a reaction that curdles the milk, creating curds.

Cutting the curd: Next, the cheesemaker cuts the curd with knives and heats it, further separating the curds and whey.

Processing the curd: Processing the curd through stirring, cooking and washing continue to acidify and dry the curds.

Draining the whey: Next, the whey is drained, leaving only a mat of cheese curds.

Cheddaring the cheese: The cheesemaker next cuts the curd mat into sections and repeatedly flips the sections before milling the mat.

Salting the cheese: For some cheeses, the next is dry salting, and for others, it is brining.

Shaping the cheese: Next, cheesemakers shape the cheese, often using molds to assist.

Aging the cheese: Some cheeses are aged for anywhere from a number of days to a number of years.

What is acetic acid fermentation?

A process of oxidation in which alcohol is converted into acetic acid by the agency of bacteria of the genus Acetobacter, especially A.aceti (as in the production of vinegar cider or wine).

Agar types that are involved in yeast and mould detection?

- ✓ Dichloran rose Bengalchloramphenicol agar (DRBC)
- ✓ Plate count agar with chloramphenicol (PCAC)
- ✓ Potato dextrose agar with chloramphenicol (PDA-C)
- ✓ Potato dextrose salt agar with chloramphenicol (PDSA-C) (for analysis of spreader moulds)

Write the methods of heating in food preservatives?

Blanching :

Blanching is steam treatment of vegetables and some fruits to fix colour and stop enzyme activities.

Pasteurisation :

Pasteurisation is a relatively mild heat treatment in which food is heated to below 100 °C.

How digestion of fats in human starts?

Digestion of fat takes place in the small intestine. Fat reaches the small intestine in the form of large globules. The liver releases bile juice which emulsifies the fat i.e. it breaks

down the large globules into smaller globules. Lipase acts and breaks down the globules into molecules.

Smoke point of milk?

Removing the milk solids increases the smoke point from 350°F to 485°F.

How yogurt is formed from milk using fermentation?

Yogurt forms when bacteria ferment the milk sugar, known as lactose, into lactic acid. The lactic acid makes the milk more acidic (lower pH) causing the proteins to coagulate.

Procedure of amylase detection in milk? 5 Marks

Amylase activity in milk is a critical quality parameter. The amylase activity in milk is often referred as diastatic power and refers to the production of reducing substances by the action of alpha & beta amylases. The measurement involves digestion of the soluble matter with a malt infusion (extract) and following increase in reducing substances by measuring reduction of Fehling's Solution. Specifically measuring alpha amylase activity (dextrinizing activity) is more complicated and is based on using a limit dextrin as substrate. Limit dextrin is prepared by action of beta amylase (free of alpha amylase activity). The beta amylase clips maltose units off the non-reducing end of the starch molecule until an alpha-1,6-brach point is encountered. The resulting product is a beta limit dextrin that serves as the substrate for the endo cleaving alpha amylase. A malt infusion is added to the previously prepared limit dextrin substrate and aliquots removed periodically to a solution of dilute iodine. The alpha amylase activity is measured by changed color of the starch iodine complex in the presence of excess beta amylase used to prepare the limit dextrin. The color is compared to a coloured disc on a comparator. This is continued until the color is matched to a color on a comparator. The time to reach that color is dextrinizing time and is a measure of alpha amylase activity, a shorter time representing a more active preparation.

Five name of dairy products? 5 Marks

- ✓ Milk
- ✓ Butter
- ✓ Cheese
- ✓ Yogurt
- ✓ Cream (heavy cream, sour cream, etc)
- ✓ Ice cream
- ✓ Paneer

Name of fruit and vegetables that contain stone?

Vegetables :

- ✓ Peas

- ✓ Beans

Fruits :

- ✓ Peaches
- ✓ Cherries
- ✓ Plums
- ✓ Apricots

Food quality regulation?

Food regulations are the legal rules that are issued by a government agency in order to implement the various food safety and food hygienic procedures in order to obtain an assurance of good and consumable quality of food product. These regulations cover the total chain beginning with reaping, handling, storage, distribution, transportation and use by the consumer.

Why Viroid cholera is effective in small number for causing food borne diseases?

In case of toxico-infection, viable cells should be consumed either in very high numbers (for those that cannot multiply in the digestive tract, such as Clo. peifringens) or in reasonable numbers (for those that can multiply in the digestive tract, such as Vibrio cholerae) so that toxins released by them in the digestive tract can produce the symptoms.

Name of bacteria in milk fermentation?

LAB (Lactic Acid Bacteria)